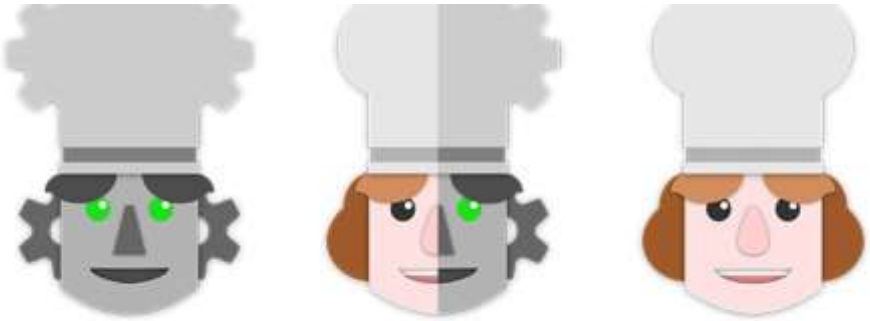
CyberChef Fundamentals

Data Manipulation, Encoding, and Extraction



CyberChef

CyberChef is a versatile, web-based tool often described as the "Swiss Army knife" of data manipulation. It enables security professionals to perform complex tasks like encoding, decoding, encryption, and data extraction through a user-friendly, "recipe-based" interface. This module details the four primary functional areas: Operations, Recipe, Input, and Output. It also outlines the critical four-step analytical workflow needed to solve data-related problems during security investigations.
By mastering categories like Extractors, Data Formats, and Time conversions, analysts can efficiently process obfuscated strings, logs, and network artifacts, making CyberChef an essential asset for both offensive and defensive cybersecurity operations.

This **documentation** and **learning** approach is strongly inspired by the structured, hands-on content provided by **TryHackMe**, which significantly contributed to practical understanding of "CyberChef Fundamentals" concepts.

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1. Introduction and Access

CyberChef is an intuitive application designed to handle "cyber" tasks within a browser. It processes data by executing a series of operations known as a **Recipe**.

Methods of Access

- **Online Access:** Accessed directly via a web browser with an internet connection.
- **Offline/Local Copy:** A stable release file can be downloaded to run locally on Windows or Linux, which is preferred for handling sensitive data without internet exposure.

2. Navigating the Interface

The CyberChef interface is divided into four distinct functional zones.

Functional Areas Overview

Area	Purpose	Key Features
Operations	The library of all available tools.	Search bar, categorized lists (e.g., Extractors, Encodings).
Recipe	The "heart" where tools are arranged.	**Bake!** button, **Auto Bake**, Save/Load recipe.
Input	Where raw data or files are placed.	Add tabs, open folder/file as input, clear input.
Output	Where the processed results appear.	Save to file, Copy to clipboard, Replace input with output.

3. The Analytical Workflow

Before processing data, a structured thought process ensures efficient results.

1. **Set Objective:** Define the goal (e.g., "I want to decode this Base64 string").
2. **Input Data:** Paste the raw, gibberish, or encoded data into the Input area.
3. **Select Operations:** Drag relevant tools into the Recipe area (e.g., **From Base64**).
4. **Check Output:** Verify if the result meets the objective. If not, refine the recipe.

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4. Common Operation Categories

CyberChef organizes its hundreds of operations into categories for easier navigation.

Extractors

Used to pull specific patterns out of a large block of messy data.

- **Extract IP addresses:** Finds all IPv4 and IPv6 instances.
- **Extract URLs:** Identifies web addresses (requires protocols like HTTP/FTP).
- **Extract email addresses:** Finds strings following the user@domain.com format.

Date and Time

Essential for correlating logs from different time zones.

- **To UNIX Timestamp:** Converts a readable date to a 32-bit value (seconds since Jan 1, 1970).
- **From UNIX Timestamp:** Converts a numerical timestamp (e.g., 1725654622) back to a readable date.

Data Format (Encodings)

- **Base64:** Transforms binary data into text.
- **Hex:** Converts strings to hexadecimal bytes.
- **URL Encode/Decode:** Handles percent-encoded characters (e.g., %20 for space).
- **ROT13:** A simple substitution cipher that rotates letters by 13 places.

5. Manual Logic: Base64 Conversion

Understanding how CyberChef works "under the hood" is vital. Base64 conversion follows these mathematical steps:

1. **Convert to Binary:** Translate characters to 8-bit binary (e.g., **T** = 01010100).
2. **Merge and Re-divide:** Combine all bits and split them into **6-bit** chunks.
3. **Decimal Conversion:** Convert the 6-bit binary chunks into Decimal numbers.
4. **Index Lookup:** Map the Decimal numbers to the **Base64 Index Table**.

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6. Practical Task (Task 6)

I researched the provided task files and **found** several encoded strings and timestamps. Below is how **I solved** the "First Official Cook":

- **Work Done: I practiced** using the **Extract IP addresses** operation on the downloaded text file. **I founded** the specific IP address that starts and ends with "10" by reviewing the output list.
- **Encoding Solution: I practiced** encoding the string "Nice Room!". **I solved** this by searching for the **To Base64** operation and **founded** the resulting encoded value.
- **URL Decoding: I researched** a long percent-encoded URL. **I practiced** using **URL Decode** and **founded** the original readable web address.
- **Timestamp Analysis: I solved** the time conversion task by dragging **From UNIX Timestamp** into the recipe. **I founded** the exact datetime string for the provided numerical value.
- **Result: I practiced** combining different "ingredients" in the recipe area and successfully retrieved all required flags.

Conclusion and Cheatsheet

Conclusion

CyberChef is an essential tool for any security professional, providing a rapid and visual way to de-obfuscate data. While it is incredibly powerful for individual artifacts, it is most effective when the analyst has a clear objective and understands the underlying data formats. Mastery of the **Recipe** area allows for the automation of complex multi-step decoding processes, saving significant time during forensic and incident response investigations.

Fast Revision Cheatsheet

- **Recipe:** A sequence of operations.
- **Bake!:** Executes the current recipe on the input.
- **Auto Bake:** Updates output in real-time as you change the recipe.
- **Extractors:** Best for finding IPs, Emails, or URLs in large logs.
- **To Hex / From Hex:** Used for analyzing binary file headers.
- **Base64:** The most common encoding for data transmission.
- **UNIX Timestamp:** Time measured in seconds since the **Epoch** (1970).
- **Magic Operation:** A special tool that attempts to automatically detect the encoding of your input.